Expanding-interval review, Anshul & Will - Final OLMo FictionalQA Review Experiment

Github: https://github.com/anshulmago1/olmo-fictionalqa-review

Research question

Does repeatedly reviewing previously learned facts help an LLM retain them while it continues learning new information? Does expanding-interval review outperform simple uniform review?

Experimental design

Crouching applicant Y a new accountModel: OLMo DataDecide 300M, approximately 377M total parameters

- **Training method:** LoRA, with 5.24M trainable parameters
- **Dataset:** FictionalQA, a dataset on fictional facts so the evaluated facts don't exist in the model's pretraining data (dataset at https://huggingface.co/datasets/jwkirchenbauer/fictionalqa)
- **Seeds:** 17, 23, and 42
- **Stage 1:** 60 updates teaching old fictional facts
- **Stage 2:** 180 updates introducing new fictional facts with reviews of old facts from stage 1 interspersed
- **Buffer:** 180 additional new-fact updates with no old-fact review
- **Evaluation:** Held-out paraphrases of the training facts
- **Review budget:** 12 review events for both review arms

The three methods of review were:

- **No review:** Old facts were never repeated.
- **Uniform:** Old facts were reviewed at approximately equal intervals.
- **Expanding:** Reviews began close together and became progressively farther apart.

The final review occurred at stage-two step 165. Including the end of stage two and the buffer, the final evaluation occurred approximately **194 updates after the last review**.

Metrics

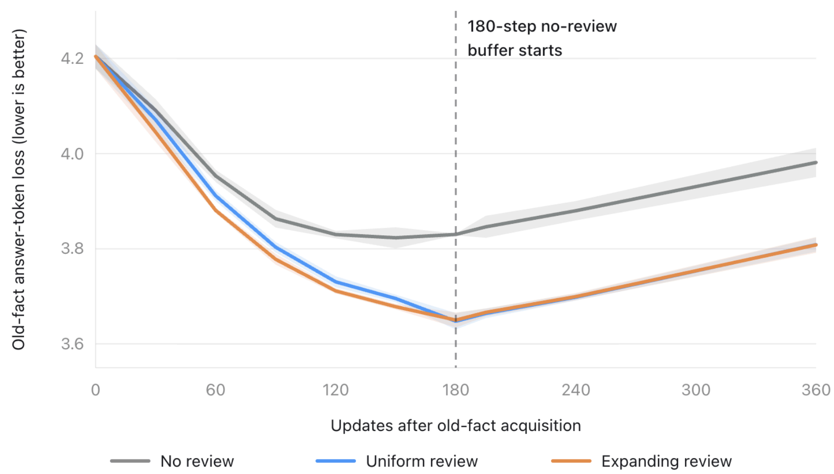
All reported losses are answer-token cross-entropy; **lower is better**.

- **Pre-buffer old loss:** Knowledge of old facts immediately before the long buffer.
- **Final old loss:** Knowledge of old facts after the 180-step buffer. This is the main delayed-retention metric.
- **Buffer forgetting:** Final old loss minus pre-buffer old loss. Positive values mean the model forgot during the buffer.
- **Old-loss change:** Final old loss minus the loss after stage one. Negative means the old facts remained better learned than at the end of stage one.
- **Trajectory change:** Average old-loss change throughout stage two. This measures how well knowledge was maintained during training, rather than only at the end.
- **New loss:** Final loss on newly introduced facts. This measures learning or plasticity.
- **Joint loss:** Equal average of final old-fact loss from stage 1 and new-fact loss from stage 2.

Main results

Values are means across three paired seeds. The ± values are standard deviations across seeds.

Condition	Reviews	Pre-buffer old loss	Buffer forgetting	Final old loss	Old-loss change	Trajectory change	New loss	Joint loss
No review	0	3.830	+0.151 ± 0.030	3.981 ± 0.031	−0.223 ± 0.021	−0.305 ± 0.014	2.993 ± 0.038	3.487 ± 0.031
Uniform	12	3.648	+0.161 ± 0.009	3.809 ± 0.015	−0.396 ± 0.015	−0.423 ± 0.021	3.018 ± 0.024	3.413 ± 0.019
Expanding	12	3.650	+0.158 ± 0.008	3.808 ± 0.016	−0.396 ± 0.014	−0.436 ± 0.023	3.015 ± 0.019	3.411 ± 0.017



Retention throughout the buffer (old fact loss, lower is better)

Buffer length	No review	Uniform	Expanding
15 updates	3.846	**3.664**	3.666
60 updates	3.880	**3.698**	3.699
180 updates	3.981	3.809	**3.808**

Both review arms maintained a clear advantage throughout the buffer compared to no review. Uniform and expanding remained almost indistinguishable at every checkpoint.

Paired comparisons

Differences are calculated as the first condition minus the second. Negative final-old-loss differences favor the first condition.

Methods compared	Final old-loss difference	95% paired CI
Uniform − no review	**−0.1727**	[−0.2140, −0.1314]
Expanding − no review	**−0.1733**	[−0.2100, −0.1366]
Expanding − uniform	−0.0006	[−0.0053, +0.0041]

Uniform reduced delayed old-fact loss by approximately **4.34%** relative to no review. Expanding reduced it by approximately **4.35%**.

The expanding-versus-uniform confidence interval is extremely narrow around zero. In this experiment, they were functionally tied.

Forgetting Rate Between Beginning and End of Buffer Period

Comparison	Difference in buffer forgetting	95% paired CI
Uniform − no review	+0.0097	[−0.0718, +0.0911]
Expanding − no review	+0.0066	[−0.0663, +0.0794]
Expanding − uniform	−0.0031	[−0.0118, +0.0057]

Every method forgot approximately **0.15–0.16 loss units** during the buffer. Review helped primarily by making the old facts better learned before the buffer, not by demonstrably changing the subsequent forgetting rate.

Retention–plasticity trade-off

This part looks at how each review method impacted how well the model retained new info that was introduced in Stage 2 and examines whether each method had an effect on retention of these new facts in addition to the old facts.

Review produced a small numerical cost on new-fact learning:

- Uniform versus no review: **+0.0252 new loss**, 95% CI [−0.0258, +0.0762]
- Expanding versus no review: **+0.0216 new loss**, 95% CI [−0.0329, +0.0760]

For the equal-weight joint metric:

- Uniform improved joint loss by **0.0737**
- Expanding improved joint loss by **0.0758**
- Expanding beat uniform by only **0.0021**, with a CI spanning zero

Conclusion

Repeated review improved long-delay factual retention by approximately **4.3%** relative to continued training without review. This advantage persisted after a substantial 180-update no-review buffer and appeared in all three paired seeds.

However, uniform and expanding review produced essentially identical final retention. Expanding performed better on the during-training trajectory metric (i.e. it performed better while training and review was ongoing), but this likely due to its front-loaded reviews. After training, there was negligible difference between uniform and expanding review.

The defensible conclusion is that:

> Under a matched review budget, periodically reviewing old facts improved their retention after prolonged continued training. Uniform and expanding review were statistically and practically indistinguishable, so the experiment supports review itself but not a specific spacing schedule.

Thus, for a production system, uniform review is the simpler default because it matches expanding review's retention without requiring a more complicated scheduler.

Andrew - P4 four-model peer distillation

Summary

This project delivers a controlled experiment for testing whether four complementary OLMo-compatible 400M peers can train a better single deployable 400M model than matched 400M students distilled from a stronger larger teacher. The primary outcome is selected `peer_frr_onpolicy` 400M minus selected `large_teacher_diverse` 400M under matched starts, data, student updates, selection policy, and sealed evaluation.

The 2-4 B200 profile is a compressed large-effect screen designed for a roughly 10-hour allocation. It is not the full 16-seed confirmatory study.

Problem

Ordinary teacher-student distillation can compress capability, but it may also overfit the student to one teacher distribution. The hypothesis is that a small population of complementary peers can expose each student to verified rescues from other peers, creating useful training signal that is harder for a single larger teacher to transfer into the same 400M deployment class.

The experiment must separate that hypothesis from easier but weaker claims, such as best-of-four ensembling, router performance, or a single greedy teacher miss.

Goals

- Test the primary peer-learning versus larger-teacher-distillation comparison with symmetric four-student selection.
- Preserve a one-model deployment endpoint, not an ensemble endpoint.
- Use the same four warmed student checkpoints across all championship arms.
- Give `large_teacher_diverse` a fair attempt budget against the peer population.
- Keep safety gates for retention, specialty preservation, shift behavior, and complementarity collapse.
- Produce auditable artifacts: manifests, per-seed results, sealed audit bundles, preaudit results, and cost ledgers.
- Support a compressed 2-4 B200 execution profile without changing the scientific arms after outcomes are visible.

Non-goals

- Proving broad general intelligence improvement.
- Claiming open-ended creativity from lexical novelty or embedding distance.
- Treating a router, ensemble, or best-of-four population result as the primary deployment claim.

- Using a 7B teacher as the primary comparator. Extreme-capacity teachers are secondary stress tests only.
- Running training without explicit operator unlock and run-mode selection.

Users and stakeholders

- Research operator: runs the notebook, supplies model and data paths, and preserves outputs.
- Team lead: approves compute spend, verifies protocol integrity, and interprets whether the result is a large-effect screen or confirmatory evidence.
- Research reviewer: inspects manifests, comparisons, safety gates, and cost accounting before accepting claims.

Required inputs

- HF-loadable OLMo-compatible 400M student checkpoint and tokenizer.
- Student checkpoint stage label via `OLMO_400M_STAGE`.
- Approximately 1B OLMo-compatible larger teacher checkpoint and tokenizer.
- Teacher stage label via `OLMO_LARGER_TEACHER_STAGE`.
- Held-out retention JSONL with one `{"text": "..."}` record per line.
- Fast writable output directory for manifests, checkpoints, and ledgers.
- Python 3.10+ environment with CUDA, PyTorch, Transformers, and NumPy.

Core arms

The championship run must include exactly these arms:

- `gold_private_equal_cost`
- `self_snapshot_op`
- `peer_frr_onpolicy`
- `large_teacher_single`
- `large_teacher_diverse`

The primary comparator is `large_teacher_diverse`, not a single greedy teacher trace.

Functional requirements

1. The notebook must remain locked unless `ALLOW_OLMO400M_TRAINING=I_UNDERSTAND_THIS_RUNS_OPTIMIZATION`.
2. The `b200_10h` profile must require an explicit `OLMO400M_RUN_MODE`.
3. `manifest_only` must create the shared deterministic data manifest before parallel seed jobs begin.

4. `championship_seed` must refuse to run under `b200_10h` if the shared manifest is missing.
5. Parallel seed jobs must write to separate `seed_<n>` directories.
6. `summarize` must refuse to summarize if any configured confirmatory seed is missing.
7. The larger teacher must pass the superiority gate before championship teacher arms train.
8. Token-level KL must be used only when tokenizer and stage compatibility make it valid; otherwise teacher arms must use verified sequence-distillation fallback.
9. The final audit must open only after all requested arms are frozen.
10. Cost ledgers must record student updates, processed tokens, auxiliary tokens, attempted outputs, decoded tokens, accepted targets, device time, and evaluation counts.

B200 compressed execution

Use this profile for a 2-4 B200, roughly 10-hour run:

export ALLOW_OLMO400M_TRAINING=I_UNDERSTAND_THIS_RUNS_OPTIMIZATION

export OLMO400M_BUDGET_PROFILE=b200_10h

export OLMO400M_B200_GPUS=4

export OLMO_400M_MODEL=/path/to/olmo_400m_student

export OLMO_400M_STAGE=<student_stage_label>

export OLMO_LARGER_TEACHER_MODEL=/path/to/olmo_1b_teacher

export OLMO_LARGER_TEACHER_STAGE=<teacher_stage_label>

export OLMO400M_RETENTION_JSONL=/path/to/retention_general_text.jsonl

export OLMO400M_EXPERIMENT_DIR=/path/to/output_dir

Run order:

1. Run `OLMO400M_RUN_MODE=manifest_only` once.
2. Run one `OLMO400M_RUN_MODE=championship_seed` process per B200.
3. Use seeds `13`, `29`, `47`, and `71` according to `OLMO400M_B200_GPUS`.
4. Run `OLMO400M_RUN_MODE=summarize` after all configured seed jobs finish.

The compressed profile should be reported as a large-effect screen. A positive mean with a failing lower-bound gate at two to four seeds should not be overinterpreted as proof of failure.

Outputs

Required output artifacts:

- `data/manifest.json`
- `model_manifest.json`
- `larger_teacher_manifest.json`
- `seed_*/seed_results.json`
- `seed_*/sealed_audit.json`
- `seed_*/*/preaudit_result.json`
- `seed_*/*/cost_events.jsonl`
- `confirmatory_summary.json`

Optional output artifact:

- `seed_*/fixed_k_novelty.json`

If the fixed-k novelty audit is omitted, success level 4 remains incomplete by design. The 10-hour core claim is the level-3 peer-vs-larger-teacher comparison.

Success metrics

Primary metric:

- `peer_vs_larger_teacher_primary_gate_passed` in `confirmatory_summary.json`.

Supporting metrics:

- paired seed analysis for `peer_frr_onpolicy` versus `large_teacher_diverse`;
- safety gate status;
- general retention margin buffers;
- expertise retention by peer;
- shift noninferiority;
- raw larger-teacher moonshot result;
- all-in cost ledger by seed and arm.

Acceptance criteria

- The notebook hash matches the recorded expected SHA-256.
- All notebook code cells parse and contain no embedded execution outputs.
- The generator compiles.
- `b200_10h` fails fast without explicit `OLMO400M_RUN_MODE`.
- `championship_seed` fails fast if the shared manifest is missing.
- `summarize` requires all configured confirmatory seeds.
- The branch contains no model weights, private retention corpus, local output directories, or generated checkpoints.

Risks

- No valid HF-loadable 400M student checkpoint is available.
- The public 1B teacher is not stage-matched to the internal 400M student, forcing sequence-distillation fallback.
- The 2-4 seed B200 run has limited statistical power and may fail the lower-bound superiority gate despite a positive mean.
- A poor retention corpus can weaken the "improved without narrowing" claim.
- The B200 profile does not enforce a wall-clock abort by itself; operators should use scheduler or process-level timeout controls.
- Outputs on ephemeral NVMe must be synced before instance shutdown.

Open questions

- What exact HF-loadable 400M checkpoint will be used?
- What stage label should be assigned to the student checkpoint?
- Is the primary teacher truly stage-matched, or should the run explicitly report sequence-distillation fallback?
- What held-out general text source will be used for retention?
- Will the optional fixed-k novelty audit be run after the 10-hour core?

Mastery Gated Learning - Adam & Katie

(mastery gating doesn't work): https://www.overleaf.com/read/fpdgpzbqkswg#ff8cfc

Claim to Prove/Disprove: Mastery learning can be used to pretrain an LLM from scratch at a fraction of the compute or the data that the mainstream LLM training pipeline uses.

Synthesis: Our report mathematically shows that mastery learning will provide a negligible speedup on the task of LLM pretraining and concludes that any gains in performance from a mastery learning setup can necessarily be explained by design choices that are *not mastery learning* (e.g. scheduling, adaptive difficulty, etc.). Training on less data is also impossible from information theory, following from the Data Processing Inequality. Moreover, Conducting any large scale experiments are infeasible for many reasons:

- Mastery is ill-defined during pretraining
- Constructing the graph itself requires a substantial effort
- Held-out eval sets leak signal into the model

Finally, while mastery learning does produce large gains in certain domains (Section 3 of the Overleaf doc), especially reinforcement learning with sparse reward, such research is far from novel and we believe that we should focus our effort elsewhere.

Verdict: Based on the research and experiments we have done, we confidently side with the null hypothesis.